

Demonstration-based Therapist Skill Transfer Learning Framework for Upper-Limb Rehabilitation Exoskeleton

Zijie Wang, Qingcong Wu, Yanghui Zhu, Jialong Wang, Cong Luo, Shen Huang, Bai Chen

Abstract— Upper limb rehabilitation exoskeletons offer a promising technological solution for neuromuscular function recovery. However, their clinical application remains challenging due to difficulties in effectively integrating therapists' professional expertise. This paper proposes a demonstration-based therapist rehabilitation skill transfer learning paradigm, where general rehabilitation assisting policy is learned from demonstration through simulation-based learning, and ultimately deployed onto a real exoskeleton system. The learning framework consists of two components: a motion imitation module that simulates muscle-actuated movements using a two-level policy, and a reinforcement learning-based exoskeleton controller that adapts assistance based on patient states. These components work jointly within a human-exoskeleton coupled system to reproduce therapist-guided movements and force patterns. Demonstration data, including video and surface electromyography, are processed to generate biomechanically consistent reference motions and torque labels. The trained controller is validated through both single-joint and multi-joint rehabilitation tasks. Experimental results show that the system can accurately replicate therapist-like motion and joint torques, and demonstrates strong adaptability across subjects and task types.

I. INTRODUCTION

Upper limb rehabilitation exoskeletons serve as a crucial technological method for neuromuscular function recovery by providing adjustable mechanical support to assist patients in restoring motor functions. Although robot-assisted rehabilitation systems have made significant advancements in motion trajectory accuracy and task adaptability in recent years [1,2], their clinical applications still face critical challenges. A major barrier is the lack of integration with therapists' expertise, particularly the real-time adjustment of assistance based on haptic interaction, which is a core aspect of conventional therapy. Clinical studies indicate that most therapists find current exoskeletons insufficient for capturing biomechanical cues derived from palpation-based assessment [1]. Moreover, a mismatch persists between interface parameters and therapists' clinical reasoning, resulting in rigid

assistance patterns that fail to meet individual rehabilitation needs [3].

In traditional rehabilitation practices, therapists dynamically perceive changes in patients' muscle tone through haptic interaction and immediately adjust the assistance force and motion trajectory accordingly. This proprioception-based decision-making process involves complex biomechanical feature extraction and motion intent prediction mechanisms. Existing studies suggest that even experienced therapists find it challenging to fully translate their professional expertise into parameterized commands executable by robots [4]. Current mainstream solutions primarily adopt pre-programmed support models or surface electromyography (sEMG)-based triggering mechanisms [5,6]. In therapist-involved approaches, real-time mapping of therapist movements and physical interaction interfaces are commonly employed [7]. However, these methods often exhibit inadequate adaptability when dealing with movement compensation, abnormal muscle tone fluctuations, and other complex conditions. More critically, current human-machine interfaces force therapists to operate within a discretized parameter space, fundamentally conflicting with their clinical decision-making processes based on continuous somatosensory feedback [1].

In recent years, learning-based methods have shown promise in various robotic control tasks [8,9]. In rehabilitation, imitation learning from therapist demonstrations or unaffected limb movements is a common approach. Motion capture data are used to learn trajectory profiles using dynamic movement primitives (DMPs) or Gaussian mixture models (GMMs) [10,11], while mirror models based on contralateral limb data support symmetric movement in hemiplegic patients [12]. However, these approaches primarily focus on motion replication and offer limited generalization in force control, making them unsuitable for real-time, patient-adaptive rehabilitation. Recently, Luo et al. [13] proposed an experiment-free method to learn a versatile control policy in simulation. Their framework demonstrated the feasibility of using simulation-trained controllers to generalize across tasks. However, integrating therapist-specific force-motion coordination into such controllers remains an open challenge, especially when relying on sparse, multimodal demonstrations.

Therefore, this paper proposes a demonstration-based therapist rehabilitation skill transfer learning paradigm for upper-limb rehabilitation exoskeleton. By constructing a demonstration dataset with muscle strength labels and integrating biomechanical simulation with DRL, the exoskeleton controller is trained to transfer the therapist's abstract assistive decision-making skills into the exoskeleton control system. This enables an online adaptive support level

*This work was supported in part by the National Natural Science Foundation of China (No. 52175014 and U22A20204), in part by the Key Research and Development Program of Jiangsu Province (Social Development) (No. BE2023813), and in part by the Open Research Fund of State Key Laboratory of Digital Medical Engineering (No. 2024-M09), and in part by the Postgraduate Research & Practice Innovation Program of NUAA (No. xcjxh20240502).

Zijie Wang, Qingcong Wu, Yanghui Zhu, Jialong Wang, Cong Luo, Bai Chen are with the College of Mechanical and Electrical Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing 210016, Jiangsu, China (corresponding author: wuqc@nuaa.edu.cn).

Shen Huang is with Biomedical Engineering Fusion Laboratory, The affiliated Jiangning Hospital of Nanjing Medical University, Nanjing 211100, Jiangsu, China.

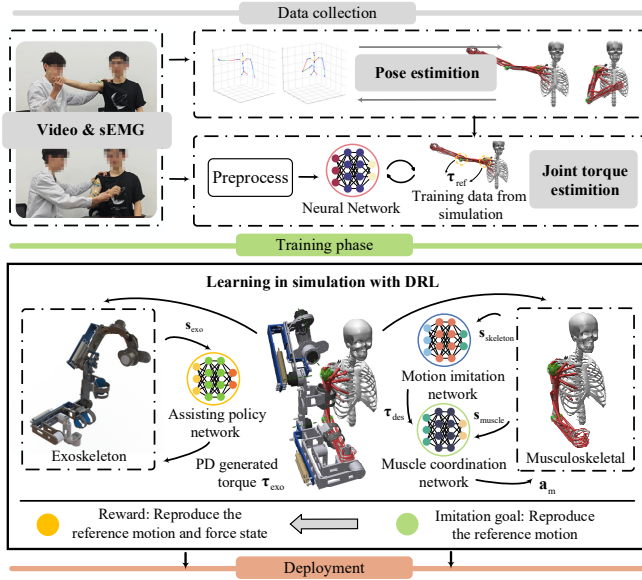


Fig. 1 Overview of the system framework

that aligns more closely with clinical expertise. The remainder of this paper is organized as follows: Section 2 introduces the overall framework and implementation of our method, including data collection, muscle-actuated upper limb motion simulation and exoskeleton assisting policy training. Section 3 presents the experimental setup and result analysis, followed by a conclusion in the final section.

II. METHOD

The system framework of the proposed method is illustrated in Fig. 1, which consists of three stages. First, data collection is performed using a monocular camera to capture the therapist-assisted rehabilitation training process while simultaneously recording the patient's sEMG signals, constructing a reference dataset labeled with muscle strength information. Next, to acquire biomechanically consistent human motor responses in a simulated environment, a low-level imitation policy is trained to actuate an upper limb musculoskeletal model to achieve the desired movements. Finally, using the reference data as a target, DRL is employed to train the exoskeleton assisting policy, enabling it to reproduce the patient's movement trajectories and force exertion patterns. The trained policy is then deployed onto a real exoskeleton platform, facilitating the digital transfer of therapist rehabilitation skills. The detailed implementation of the proposed method is described in the following sections.

A. Data Collection from Demonstration

To construct the demonstration dataset, we record rehabilitation sessions where a therapist assists a patient through typical upper-limb tasks. The video and sEMG signals captured in this process inherently contain both patient responses and therapist-guided interventions, representing therapist movement indirectly through the mechanical interactions observed in joint torque and motion profiles. Although we do not explicitly segment therapist motion data, our framework leverages the composite movement outcome generated through therapist-patient interaction, enabling effective learning of the therapist-style assistance. As therapist-applied force primarily affects the shoulder and

elbow joints [14], we focus on four degrees of freedom (DOFs) in these regions.

To effectively characterize both isolated motor control ability and coordinated activation patterns, we collect sEMG signals from eight key muscles in the upper limb kinetic chain. For the shoulder joint, the anterior, middle, and posterior deltoid, supraspinatus, infraspinatus, and pectoralis major are selected. For the elbow joint, the biceps brachii and triceps brachii are chosen, covering the primary driving and stabilizing structures in both single- and multi-joint movement patterns. It should be noted that we do not directly record the therapist's sEMG signals since the primary control objective is to reproduce the effect of therapist intervention on the patient's limb. The patient's sEMG during assisted sessions reflects both voluntary effort and therapist-induced responses, allowing our model to learn a force-motion coupling strategy that captures therapist influence while adapting to patient states in real time.

Then, neural network is employed to construct a nonlinear model for estimating joint torques. In our method, the exoskeleton controller is trained in a simulated environment; therefore, the predicted joint torque data should be referenced against environmental variables in simulation. Here, we use the kinematic data extracted in step (a) to simulate reference joint torques on an upper limb musculoskeletal model. These torque data are then used as training data for the network to establish a connection between sEMG signals and force information that cannot be directly simulated. Our network takes the preprocessed sEMG signals as input and outputs the corresponding reference joint torques. The model is trained using backpropagation to iteratively minimize the mean squared error between predicted and reference torques, allowing the network to learn the nonlinear mapping from sEMG signals to joint torque outputs.

From the video, 3D human poses are extracted in SMPL format [15]. YOLOv4 [16] is used for subject tracking, followed by ViTPose [17] for 2D keypoint detection, and HybrIK [18] to estimate SMPL parameters. To match the MyoSim musculoskeletal model [19], we map SMPL joints to biomechanically relevant joint angles (shoulder and elbow) via inverse kinematics, downsampling and normalizing them to serve as simulation inputs.

B. Muscle-Actuated Upper Limb Motion Simulation

We simulate human motion using the MyoSim upper-limb musculoskeletal model [19], which includes a hinge elbow joint, a ball-and-socket shoulder joint, and 63 muscle-tendon units (Fig. 1). Muscle contractions generate motion by pulling on bones, and the resulting joint torques are computed via inverse dynamics using the musculoskeletal Jacobian, serving as reference for training and evaluation.

To ensure biologically realistic motion generation, we adopt the two-level policy framework described in [20] to train the controller of the musculoskeletal model, enabling it to produce motion similar to the reference data. At the higher level, the motion imitation network follows a stochastic policy $\pi_\theta(\mathbf{u} | \mathbf{s}_{\text{skeleton}})$, generating the target posture $\mathbf{u}$ for the next time step as the output based on the current skeletal state $\mathbf{s}_{\text{skeleton}}$. Proportional-derivative (PD) controller is then used

to generate the desired acceleration, which is then passed to the policy of the lower level. Here, θ represents the network parameters optimized using DRL. The lower-level muscle coordination network generates a deterministic policy $\pi_{\psi}(\dot{\mathbf{q}}_{des}, \mathbf{s}_{muscle})$ that activates muscles to produce the desired motion, with ψ representing the network parameters determined by regression. The muscle coordination policy, conditioned on skeletal kinematics and dynamics, drives muscle activations that biomechanically influence skeletal states, forming closed-loop dynamics. This hierarchical architecture employs a high-level policy encoding motion intent via target postures and a low-level policy enforcing musculoskeletal compliance through physiological muscle activations. The decomposition aligns with human motor control hierarchies, separating task planning from biomechanical execution to facilitate efficient DRL optimization.

The state of the musculoskeletal model is defined as $\mathbf{s}_{skeleton} = (\mathbf{q}_t, \dot{\mathbf{q}}_t)$, where $\mathbf{q}_t$ and $\dot{\mathbf{q}}_t$ represent the set of joint angles and angular velocities at time t . The state vector is 8-dimensional. The reward function encourages the agent to mimic the reference motion and produce smooth skeletal movement, defined as follow:

$$r_t = w^p r_t^p + w^v r_t^v + w^{as} r_t^{as} + w^{reg} r_t^{reg} \quad (1)$$

where $r_t^p, r_t^v, r_t^{as}, r_t^{reg}$ represent the position reward, velocity reward, trajectory smoothness reward, and velocity regularization reward, respectively. The first two rewards are computed by calculating the MSE with respect to the reference data, followed by negative exponential normalization.

$$\begin{aligned} r_t^p &= \exp(-\lambda^p e_t^p), \quad e_t^p = \text{MSE}(\mathbf{q}_t, \hat{\mathbf{q}}_t), \\ r_t^v &= \exp(-\lambda^v e_t^v), \quad e_t^v = \text{MSE}(\dot{\mathbf{q}}_t, \hat{\dot{\mathbf{q}}}_t) \end{aligned} \quad (2)$$

The trajectory smoothness reward encourages the agent to generate smooth trajectory curves. Here, we impose a penalty term to constrain the expected acceleration during model motion.

$$r_t^{as} = \exp(-\alpha_{as} \|(\ddot{\mathbf{q}}_{des})_t - 2(\ddot{\mathbf{q}}_{des})_{t-1} + (\ddot{\mathbf{q}}_{des})_{t-2}\|^2) \quad (3)$$

Finally, the velocity regularization term is used to suppress high-frequency oscillations, and it is computed as follows:

$$r_t^{reg} = \exp(-\lambda^{reg} e_t^{acc}), \quad e_t^{acc} = \text{mean} \left(\frac{\|\ddot{\mathbf{q}}_t\|^2}{\|\hat{\dot{\mathbf{q}}}_t\|^2 + \lambda^{reg}} \right) \quad (4)$$

where λ^{reg} is a hyperparameter used to adjust sensitivity; $\ddot{\mathbf{q}}_t$ represents the joint acceleration of the simulated musculoskeletal model; and $\hat{\dot{\mathbf{q}}}_t$ denotes the reference joint velocity.

We use the Proximal Policy Optimization (PPO) algorithm [21] to update the motion imitation network, as it has demonstrated exceptional performance in a range of high-dimensional control tasks. To stabilize value function learning, the model is trained using a combination of multi-step returns and temporal-difference learning. Additionally, generalized advantage estimation (GAE) is

used to reduce variance and improve the stability of policy gradient estimation.

The muscle coordination network determines the activation levels of all muscles to achieve the desired joint acceleration. Since the human body has more muscles than the minimum required to achieve joint motion, muscle coordination is an underdetermined problem with an infinite number of possible solutions. We formulate this as a regression problem within a supervised learning framework, allowing it to be integrated into the DRL framework. The training loss function is defined as:

$$\text{Loss}(\mathbf{a}(\psi)) = \mathbb{E} \left[\|\tilde{\mathbf{q}}_d(\mathbf{u}) - \mathbf{L}\mathbf{a}(\psi) - \mathbf{b}\|^2 + w_{reg} \|\mathbf{a}(\psi)\|^2 \right] \quad (5)$$

$$\mathbf{L} = \mathbf{M}^{-1} \mathbf{A}, \mathbf{b} = \mathbf{M}^{-1} (\sum_m \mathbf{J}_m^T \mathbf{f}_m(0) + \mathbf{J}_c^T \mathbf{f}_c + \boldsymbol{\tau}_{ext} - \mathbf{c}) \quad (6)$$

where $\mathbf{M}$ is the mass matrix, $\mathbf{a}_m(\psi)$ is the muscle activation vector, $\mathbf{J}_m$ and $\mathbf{J}_c$ are the Jacobian matrices that map forces to generalized coordinates. $\mathbf{f}_m$ and $\mathbf{f}_c$ represent muscle forces and constraint forces. $\mathbf{c}$ denotes Coriolis and gravitational forces. A detailed derivation of this process can be found in [20].

C. Exoskeleton Assisting Policy Learning

In our previous work, we developed an upper limb rehabilitation exoskeleton [22], which serves as the research platform for this study (Fig. 1). The exoskeleton features seven degrees of freedom, supporting movements of the shoulder, wrist, and elbow joints. Its workspace accommodates most upper limb functional tasks. In our approach, the exoskeleton controller is trained based on the musculoskeletal motion. To be specific, the motion imitation policy is trained firstly using a musculoskeletal model. Then, after incorporating the exoskeleton constraints, the exoskeleton assistance policy is further trained. Given the minimal interaction force between the exoskeleton and the arm, as well as the constraints of the straps and end-effector, their relative displacement is negligible. Therefore, in the simulation, their connection is not explicitly modeled, and a fixed connection is assumed at the strap attachment points.

It is important to note that the patient is not assumed to be completely passive. Instead, real-time sEMG data is used to estimate active torque contribution, and the exoskeleton must adaptively adjust its support based on this. Thus, to enable policy adaptation to varying patient effort levels, dynamic muscle fatigue, and task-specific force patterns, we use DRL to train the exoskeleton controller as well. The exoskeleton assisting policy network takes the proprioceptive state of the human body as inputs, outputs a target trajectory curve for each joint of the exoskeleton, and then generates the actuation torque via a PD controller. The objective of learning is to ensure that the human model, assisted by the exoskeleton, closely matches the reference data in terms of joint angle and torque. To generate smoother and more natural movements while reducing task complexity to accelerate policy convergence, we construct the target trajectory profile incrementally. The input state of the network is defined as $\mathbf{s}_{exo} = ((\mathbf{q}^{exo})_t, (\dot{\mathbf{q}}^{exo})_t, (\mathbf{q}^{exo}_{des})_t, \boldsymbol{\tau}_t)$, where $\mathbf{q}^{exo}, \dot{\mathbf{q}}^{exo}$ represent the exoskeleton joint angles and velocity vectors, respectively. $(\mathbf{q}^{exo}_{des})_t$ denotes the target position at the current time step, and

τ_t represents the human joint torque. To replicate the reference motion state, the reward is computed as follows:

$$r_{t,exo} = w_{exo}^p r_{t,exo}^p + w_{exo}^v r_{t,exo}^v + w^{\tau} r_t^{\tau} \quad (7)$$

where $r_{t,exo}^p, r_{t,exo}^v$ represent the position and velocity rewards, respectively, both of which are calculated using Equation (2). r_t^{τ} is the force state reward used to evaluate the difference between the human force output under exoskeleton assistance and the reference motion. It is computed using the following equation:

$$r_t^{\tau} = \exp(-\lambda^{\tau} e_t^{\tau}), \quad e_t^{\tau} = \text{MSE}(\tau_t, \hat{\tau}_t) \quad (8)$$

where τ_t represents the joint torque of the human body at time t , $\hat{\tau}_t$ is the reference torque, and λ^{τ} is the corresponding hyperparameter used to control the sensitivity of this term. Subsequently, to further adjust the joint torque response, a PD controller is used to convert the target trajectory profile into the driving torque:

$$\tau_{exo} = k_p(\mathbf{q}_{des}^{exo} - \mathbf{q}^{exo}) - k_v \dot{\mathbf{q}}^{exo} \quad (9)$$

where $\mathbf{q}_{des}^{exo}$ represents the target position, and k_p, k_v are the proportional and derivative coefficients, respectively.

To facilitate the transfer of the trained controller from simulation to the real world, domain randomization is employed. During the training process, we perturb the exoskeleton and musculoskeletal dynamics parameters (mass, inertia matrix, center of mass position, and joint friction and damping coefficients) to enhance the robustness of the policy. Additionally, considering the variation in muscle strength among different wearers, the maximum isometric force in the Hill-type muscle model is also perturbed within a specified range to improve the generalization ability of the policy.

III. EXPERIMENTS AND RESULTS

A. Experimental Setup

The exoskeleton system (Fig. 2) has seven degrees of freedom, with each rotary joint actuated by servo motors, featuring RD-X8-S2 motor (Myactuator, China) at J1, RD-X6-S2 motor (Myactuator, China) at J2/J3/J4, and DM-J4310-2EC motor (dmBot, China) at J5/J6/J7. The mechanical structure features adjustable link lengths to accommodate different users. Before each trial, the exoskeleton was manually fitted to align with the subject's upper-limb dimensions. sEMG signals were recorded in real time using a wireless collector (gForceOct, OYMTion) and used for online joint torque estimation, while the assisting policy generated adaptive support trajectories based on the participant's motion state. Control and sampling frequencies were both set to 1000 Hz. Participants were required to wear the exoskeleton on their right arm, follow a demonstration video and execute the guided movements.

To validate the effectiveness of the proposed method, two sets of experiments were conducted to evaluate the performance of the exoskeleton controller in rehabilitation assistance tasks, as illustrated in Fig. 3. The first set involves four single-joint training task: shoulder flexion-extension (S-FE), internal-external rotation (S-IR/ER), abduction-adduction (S-AA), and elbow flexion-extension

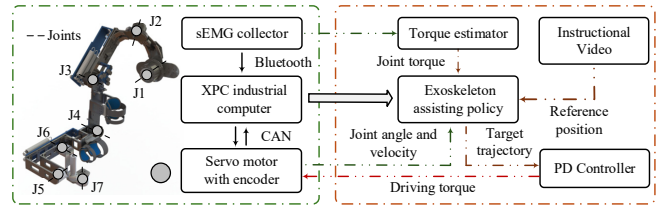


Fig. 2 Control system of the upper limb rehabilitation exoskeleton.

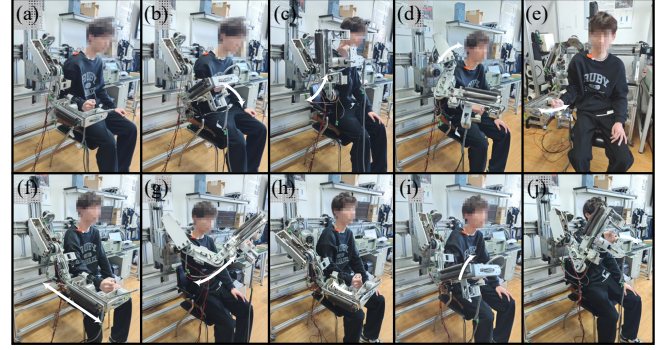


Fig. 3 Experiment tasks. (a) Resting Pose. (b) Elbow flexion-extension. (c) Shoulder abduction-adduction. (d) Shoulder flexion-extension. (e) Shoulder internal-external rotation. (f) Pushing-pulling. (g) Diagonal delivery. (h)-(j) Air circular drawing.

(E-FE), targeting the prevalent abnormal synergy patterns in early-stage rehabilitation patients. The second set includes three multi-joint tasks: push-pull (PP), diagonal delivery (DD), and air circular drawing (ACD), designed for functional task reconstruction in mid-to-late-stage rehabilitation patients. The experiment involved five healthy subjects (S1-S5; mean±SD: age 24.2±1.5 years, weight 68.2±5.1 kg, height 175.3±4.2 cm). Each movement was performed in three cycles and repeated three times. The experimental protocol was approved by the Institutional Review Board of Nanjing University of Aeronautics and Astronautics.

B. Results and Discussion

In the single-joint isolated movement experiment, taking participant S1 as an example, the comparison between the experimental data and the demonstration data is shown in Fig. 4. The subfigures (a)–(d) illustrate the comparison of joint angles between four experimental groups and the reference data collected from the demonstration. “Actual” represents the mean value of three repeated experiments, with the shaded area indicating the error band. Subfigures (e)–(h) show the comparison between the torques estimated from the subject's sEMG signals and the torques extracted from the demonstration. “Predicted” represents the mean value of three repeated experiments, with the shaded area also indicating the error band. To quantitatively evaluate the controller's performance, the root mean square error (RMSE) for each task was calculated, as shown in Table 1. The angular tracking error show minimal variation across the four degrees of freedom, with the S-IR/ER exhibiting the highest tracking accuracy (relative error reduction of 14.1% to 69.7%). This phenomenon is preliminarily hypothesized to be related to the disturbance of the controller caused by spring oscillations in the exoskeleton's gravity compensation mechanism. The joint torque tracking error exhibit greater dispersion, but its overall trend corresponds to that of the demonstration data.

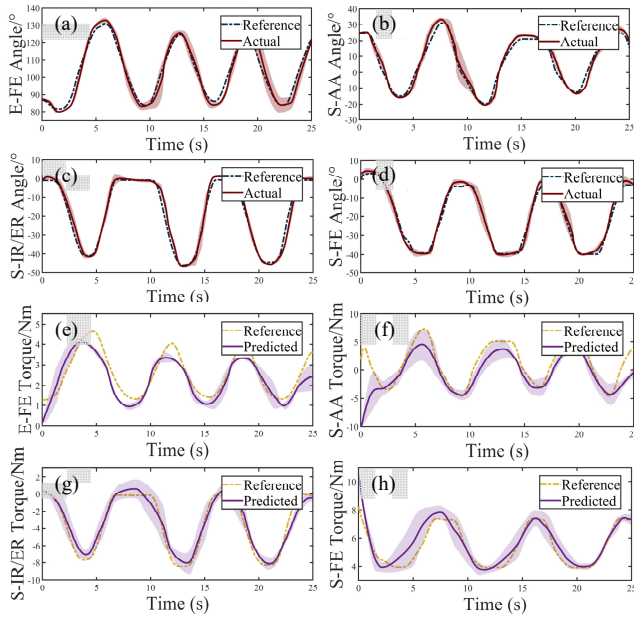


Fig. 4 Joint angle and torque in single-joint isolation experiment. (a)-(d) Reference and actual angles of elbow and shoulder. (e)-(h) Reference and predicted torque of elbow and shoulder.

The maximum absolute error (AAE) of the elbow joint torque is 0.417 Nm, which is below the torque perception threshold defined in [23]. The AAE of the three degrees of freedom of the shoulder joint is 0.922 Nm. As a multi-DOF composite joint, the torque perception threshold of the shoulder may vary significantly across different movement planes, and little quantitative analysis has been conducted on this aspect so far. However, according to participant feedback, the perception of the exoskeleton-assisted torque during the experiment did not differ significantly from that of the therapist.

TABLE I. RMSE OF JOINT ANGLE AND TORQUE FOR TASKS IN SINGLE-JOINT ISOLATION EXPERIMENT

	E-FE	S-AA	S-IR/ER	S-FE
RMSE of Angle/ $^{\circ}$	2.936	4.366	2.573	3.566
RMSE of Torque/Nm	0.585	0.545	0.801	0.685

In the multi-joint coordinated movement experiment, taking ACD as an example, the comparison between the joint angle and torque data of participant S1 and the demonstration data is shown in Fig. 5, while the statistical data of RMSE is listed in Table 2. Fig. 4 (a)-(d) illustrate the comparison of four joint angles between the ACD experiment and the reference data collected from the demonstration, while (e)-(h) show the comparison between the torques estimated from the subject's sEMG signals and the torques extracted from the demonstration. Compared to single-joint isolated movements, the error in multi-joint coordinated motion is significantly larger, which aligns with our expectations. The increased modeling error of multi-DOF coupled dynamics contributes to greater uncertainty in the movement process. And part of the joint torque tracking error may also be attributed to the intrinsic variability and noise of sEMG signals, which are known to suffer from cross-talk, electrode placement sensitivity, and muscle fatigue effects. In terms of joint angles, the average RMSE (ARMSE) across the four degrees of

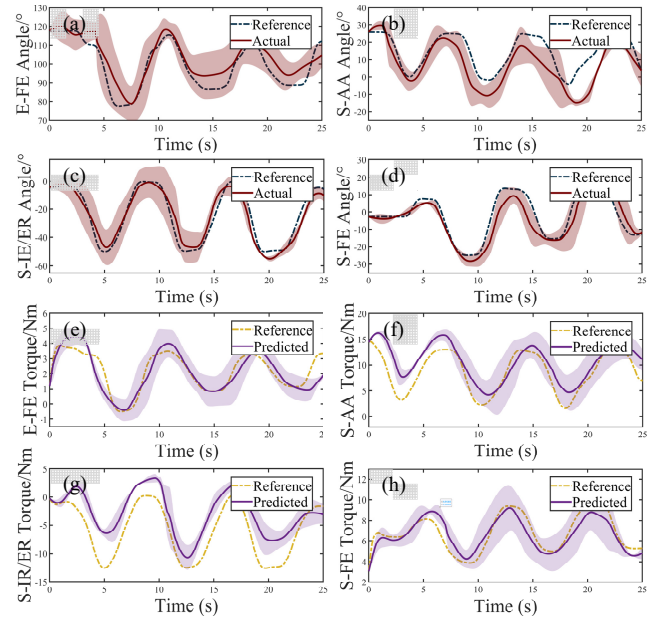


Fig. 5 Joint angle and torque in task of air circular drawing. (a)-(d) Reference and actual angles of elbow and shoulder. (e)-(h) Reference and predicted torque of elbow and shoulder.

freedom is 8.486° , while for joint torques, it is 2.53 Nm. The variation trends remain consistent with the reference data. In PP and DD tasks, the ARMSE of joint angles is 9.014° and 10.368° , while that of joint torques is 1.058 Nm and 3.807 Nm. The preliminary experimental results indicate that the trained controller can effectively reproduce the demonstrated motions across various training tasks. Meanwhile, the joint torque levels of participants exhibit a high degree of consistency with the reference data.

TABLE II. RMSE OF JOINT ANGLE AND TORQUE FOR TASK OF AIR CIRCULAR DRAWING

	E-FE	S-AA	S-IR/ER	S-FE
RMSE of Angle/ $^{\circ}$	13.731	12.862	4.356	2.995
RMSE of Torque/Nm	1.071	3.419	4.303	1.327

To evaluate the adaptability of our method to different wearers, we compared the experimental data of five subjects, as shown in Figure 6. Subfigures (a) and (b) present the statistical distribution of joint angle and torque errors for different subjects in the single-joint experiment, while subfigures (c) and (d) show the corresponding distributions for the multi-joint tasks. In the single-joint isolated movement experiment, the ARMSE for joint angles across four experimental movements among the five subjects is 1.007° , while that for torques is 0.261 Nm. In the multi-joint coordinated movement experiment, the ARMSE for joint angles across three experimental movements is 3.404° , while that for torques is 0.286 Nm. It should be noted that the trained policy is designed to be subject-agnostic and is shared across all participants. The model is trained in simulation using diverse muscle and kinematic parameters (via domain randomization), enabling generalization across subjects with different body shapes and motion patterns. Both the single-joint and multi-joint experiments use the same trained

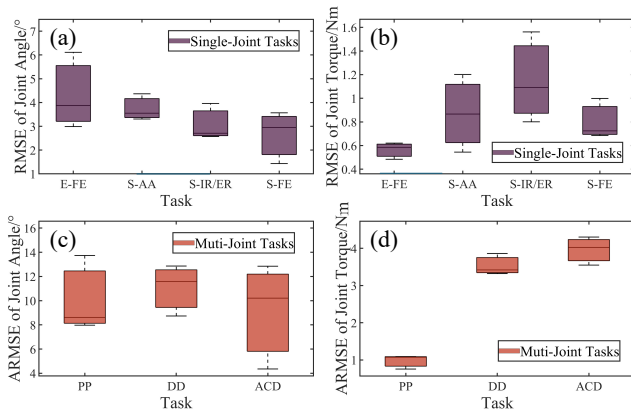


Fig. 6 RMSE statistics in five subjects on different tasks. (a)-(b) The statistical distribution of joint angle and torque RMSE for different subjects in single-joint tasks. (c)-(d) The statistical distribution of joint angle and torque ARMSE for different subjects in multi-joint tasks.

policy without retraining. The results indicate that our method demonstrates reliable adaptability to different wearers, exhibiting stable performance consistency across both single-joint and multi-joint movement patterns. Individual differences among subjects, such as body shape and movement habits, did not significantly interfere with the system, confirming the robustness of the algorithm in complex wearable scenarios.

IV. CONCLUSION

This paper proposes a demonstration-based rehabilitation skill learning method that integrates therapists' decision-making abilities into the exoskeleton control system, providing real-time and effective support for patients. The training process mainly consists of two parts: human motion imitation and exoskeleton assisting policy learning. Human motion is modeled using a two-level policy framework to simulate muscle-actuated upper limb motions, while the exoskeleton assisting policy is trained based on DRL framework. The two components work collaboratively to reproduce the demonstrated state of patients in the human-exoskeleton coupled system. Experiments on real exoskeleton platform demonstrate that the system effectively reproduces the motions and joint torque in demonstration, confirming the feasibility of the learning-based controller in therapist-involved exoskeleton systems. However, we acknowledge that transitioning between early- and late-stage rehabilitation, where patient capabilities and therapist strategies vary, may necessitate model refinement or partial retraining. In future work, stage-specific policy adaptation or progressive learning strategies will be explored to better match dynamic patient needs.

REFERENCES

- [1] C. Celian, V. Swanson, M. Shah, C. Newman, B. Fowler-King, et al., "A day in the life: a qualitative study of clinical decision-making and uptake of neurorhabilitation technology," *J. Neuroeng. Rehabil.*, vol. 18, p. 121, 2021.
- [2] Q. Wu, Z. Wang, Y. Chen, and H. Wu, "Barrier Lyapunov function-based fuzzy adaptive admittance control of an upper limb exoskeleton using RBFNN compensation," *IEEE/ASME Trans. Mechatronics*, vol. 30, no. 1, pp. 3-14, 2025.
- [3] D. Lapidou, F. Curtis, J. Akanuwa, K. Goher, A. N. Siriwardena, and A. Kucukilmaz, "Patient, carer, and staff perceptions of robotics in

- motor rehabilitation: A systematic review and qualitative metasynthesis," *J. Neuroeng. Rehabil.*, vol. 18, no. 1, p. 181, 2021.
- [4] F. Just, *Fusing Conventional and Robotic Rehabilitation Therapy*, Ph.D. dissertation, Dept. Health Sci. Technol., Eidgenössische Technische Hochschule (ETH), Zurich, Switzerland, 2020.
- [5] Q. Wu, Z. Wang and Y. Chen, "sEMG-based adaptive cooperative multi-mode control of a soft elbow exoskeleton using neural network compensation," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 31, pp. 3384-3396, 2023.
- [6] Q. Wu and Y. Chen, "Adaptive cooperative control of a soft elbow rehabilitation exoskeleton based on improved joint torque estimation," *Mech. Syst. Signal Process.*, vol. 184, p. 109748, 2023.
- [7] S. Massardi, D. Pinto-Fernandez, J. Babič, M. Dežman, A. Trošt, et al., "Learning by demonstration for motion planning of upper-limb exoskeletons," *Front. Neurobot.*, vol. 12, p. 5, 2018.
- [8] D. Han, B. Mulyana, V. Stankovic, and S. Cheng, "A survey on deep reinforcement learning algorithms for robotic manipulation," *Sensors*, vol. 23, no. 7, p. 3762, 2023.
- [9] Y. Zhu, Q. Wu, B. Chen, Q. Tang, H. Zeng and H. Wu, "Design and torque control of a modular reconfigurable variable stiffness exoskeleton for improved transparency," *IEEE/ASME Trans. Mechatron.*, pp. 1-12, 2025.
- [10] N. Pavon-Pulido, J. Lopez-Riquelme, and J. Feliu-Batlle, "IoT architecture for smart control of an exoskeleton robot in rehabilitation by using a natural user interface based on gestures," *J. Med. Syst.*, vol. 44, p. 144, 2020.
- [11] Q. Wu, L. Zheng, Y. Zhu, Z. Xu, Q. Zhang, and H. Wu, "Development of a reconfigurable 7-dof upper limb rehabilitation exoskeleton with gravity compensation based on DMP," *IEEE Trans. Med. Robot. Bionics*, vol. 7, no. 1, pp. 303-314, 2025.
- [12] B. Strong, B. Zeng, P. McCarthy, A. Roula and L. Guo, "A framework to design virtual reality mirror therapy (VRMT) for motor rehabilitation in post-stroke survivors: dosage, motivation, task difficulty, feedback and mechanism," in *Proc. IEEE GEM Conf.*, Turin, Italy, 2024, pp. 1-6.
- [13] S. Luo, M. Jiang, S. Zhang, J. Zhu, S. Yu, et al., "Experiment-free exoskeleton assistance via learning in simulation," *Nature*, vol. 630, pp. 353-359, 2024.
- [14] B. Luciani, M. Sommerhalder, M. Gandolla, P. Wolf, F. Braghin, and R. Riener, "Therapists' force-profile teach-and-mimic approach for upper-limb rehabilitation exoskeletons," *IEEE Trans. Med. Robot. Bionics*, vol. 6, no. 4, pp. 1658-1665, 2024.
- [15] M. Loper, N. Mahmood, J. Romero, G. Pons-Moll, and M. J. Black, "SMPL: A Skinned Multi-Person Linear Model," *ACM Trans. Graph.*, vol. 34, no. 6, Art. no. 248, 2015.
- [16] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "YOLOv4: optimal speed and accuracy of object detection," arXiv:2004.10934, 2020. [Online]. Available: <https://arxiv.org/abs/2004.10934>
- [17] Y. Xu, J. Zhang, Q. Zhang, and D. Tao, "ViTPose++: Vision transformer for generic body pose estimation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 2, pp. 1212-1230, 2024.
- [18] J. Li, C. Xu, Z. Chen, S. Bian, L. Yang, and C. Lu, "HybRIK: A hybrid analytical-neural inverse kinematics solution for 3d human pose and shape estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 3383-3393.
- [19] H. Wang, V. Caggiano, G. Durandau, M. Sartori, and V. Kumar, "MyoSim: Fast and physiologically realistic mujoco models for musculoskeletal and exoskeletal studies," in *Proc. IEEE Int. Conf. Robot. Automat. (ICRA)*, Philadelphia, PA, USA, 2022, pp. 8104-8111.
- [20] S. Lee, M. Park, K. Lee, and J. Lee, "Scalable muscle-actuated human simulation and control," *ACM Trans. Graph.*, vol. 38, no. 4, Art. no. 73, 2019.
- [21] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," arXiv:1707.06347, 2017. [Online]. Available: <https://arxiv.org/abs/1707.06347>
- [22] L. Zheng, Q. Wu, Y. Zhu and Q. Zhang, "Design and evaluation of a reconfigurable 7-dof upper limb rehabilitation exoskeleton with gravity compensation," in *Proc. IEEE Int. Conf. Robot. Automat. (ICRA)*, Yokohama, Japan, 2024, pp. 190-196.
- [23] H. Kim and A. T. Asbeck, "Just noticeable differences for elbow joint torque feedback," *Sci. Rep.*, vol. 11, no. 1, 2021.